

Acoustics of a Liquid with Droplets Covered by a Shell in the Presence of Phase Transitions

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Abstract—The modified Rayleigh–Lamb equation that takes into account radial oscillations of a liquid droplet covered by a viscoelastic shell and located in the viscoelastic medium is derived. At the center of the droplet there is a vapor bubble. The problem of heat- and mass transfer between the vapor, the liquid phase, the shell, and the carrying liquid is solved for small oscillations of inclusion. The energy equation is found. The dispersion relation is derived based on the obtained equations for radial oscillations of inclusion, the energy equation, and the well-known wave equation for a bubbly liquid. The dependences of the phase velocity and the attenuation coefficient on the perturbation frequency are constructed for a mixture of liquid with dodecafluoropentane C_5F_{12} and octafluoropropane C_3F_8 droplets covered by a shell with vapor bubbles at the center. The effect of phase transitions, shell of inclusions, and elastic carrying phase on dynamics of acoustic waves is illustrated. In a particular case, the theory is compared with the well-known experimental data.

Keywords: acoustic waves, liquid droplet, vapor bubble, viscoelastic shell, mass transfer

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In recent times the study of emulsions with phase transitions in which vapor bubbles are formed inside liquid droplets under the impact of ultrasound is of great interest. This process is known as acoustic droplet vaporization [1]. The use of such emulsions has a vast potential in biomedicine [2, 3] since, as distinct from usual gas microbubbles, the liquid droplets can be fairly small and can quietly flow through thin capillaries. For example, when using sonographic visualization, the liquid droplets, that flow through the vascular system of tumors, can evaporate and be used as the contrast agents to obtain images of the internal tumor structure [4]. As the liquid droplets, perfluorocarbon droplets such as decafluorobutane C_4F_{10} , octafluoropropane C_3F_8 and dodecafluoropentane C_5F_{12} droplets are usually used [5–7]. The choice of such matters is connected with the fact that these matters remain metastable at the physiologic temperature and cannot evaporate spontaneously. An acoustic energy is necessary to evaporate them. Usually, in order to prevent rapid droplet dissolution in the carrying liquid the droplets are covered by a polymeric shell [8].

In [9] the analytic solution to the problem of droplet vaporization was first obtained. In that study the Rayleigh–Plesset equation that describes the evolution of bubbles inside a superheated perfluorocarbon droplet located at the center of rigid pipe was derived. However, this model was obtained under a strong simplification, namely, the droplet vaporization rate was considered to be constant during the entire evaporation process. In [10, 11] heat transfer and the vaporization rate were described as functions of the temperature based on the approach used in simulation of a vapor bubble in usual liquid [12]. The numerical calculations were compared with superfast optic measurements of the vapor-gas bubble growth. Important role of gas diffusion in the growth of inclusion was underlined. However, both models were obtained under the assumption that the vapor bubble is located in the infinite perfluorocarbon liquid medium. In [13] a mathematical model of dynamics of growth of a vapor bubble located inside a liquid droplet of finite dimension was proposed. It was found that the theoretical calculations of variation in the vapor bubble radius of C_5F_{12} , C_3F_8 , and C_4F_{10} droplets located in water and the experimental data are in good agreement. In [14, 15] the viscoelastic droplet shell was taken into account. It was shown that the presence of

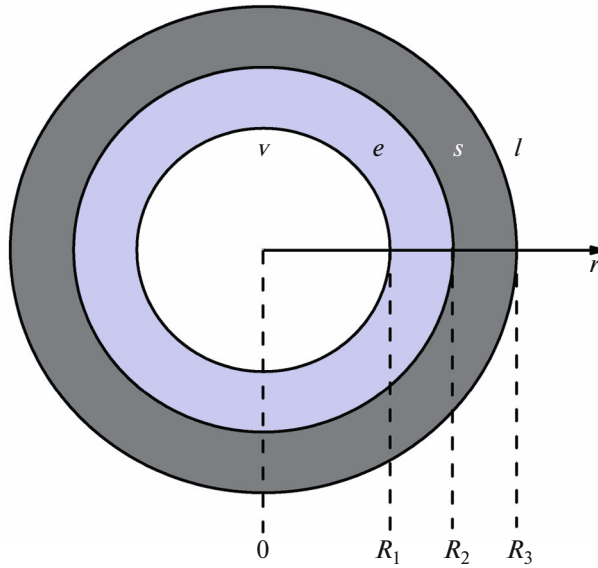


Fig. 1. Diagram of an inclusion: v denotes the vapor, e denotes the liquid phase, s is the shell, and l is the carrying liquid.

the elastic shell has the significant effect on the process of droplet vaporization. In [16, 17] the influence of the rheological properties of carrying liquid on bubble dynamics was investigated. In the present study, acoustic wave dynamics in the viscoelastic medium with liquid droplets covered by an elastic shell inside which vapor bubbles are located is first considered.

1. EQUATION OF RADIAL OSCILLATIONS

We will consider a liquid droplet covered by viscoelastic shell with a vapor bubble at the center. The droplet is located in the external viscoelastic liquid (Fig. 1). In the presence of central symmetry, the continuity and momentum equations can be written in the spherical coordinate system as follows:

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_v \rho_v) = -\frac{\partial \rho_v}{\partial t}, \quad r < R_1, \quad (1.1)$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 u_k) = 0, \quad r > R_1, \quad k = e, s, l, \quad (1.2)$$

$$\rho_k \left(\frac{\partial u_k}{\partial t} + u_k \frac{\partial u_k}{\partial r} \right) = -\frac{\partial p_k}{\partial r}, \quad r > R_1, \quad k = e, s, l. \quad (1.3)$$

Here, ρ is the density, r is the radial coordinate, t is time, u is the radial velocity, and p is the pressure. The solutions of Eqs. (1.1) and (1.2) take the form:

$$u_v = \frac{r}{R_1} U_v, \quad u_k = \frac{R_1^2}{r^2} U_k, \quad k = e, s, l, \quad (1.4)$$

where U_v and U_k are the vapor, liquid, and shell velocities on the boundaries of inclusion, respectively. It should be noted that only uniform deformation of vapor is considered.

We can write the boundary conditions on the interfaces R_1 with regard to the phase transitions and on R_2 and R_3 without regard for phase transitions

$$p_e(R_1) - p_v(R_1) + \frac{2\sigma_1}{R_1} = \tau_e(R_1) - \tau_v(R_1) + J(u_e(R_1) - u_v(R_1)), \quad (1.5)$$

$$J = \rho_v(\dot{R}_1 - U_v) = \rho_e(\dot{R}_1 - U_e),$$

$$p_s(R_2) - p_e(R_2) + \frac{2\sigma_2}{R_2} = \tau_s(R_2) - \tau_e(R_2), \quad u_e(R_2) = u_s(R_2) = \dot{R}_2, \quad (1.6)$$

$$p_l(R_3) - p_s(R_3) + \frac{2\sigma_3}{R_3} = \tau_l(R_3) - \tau_s(R_3), \quad u_s(R_3) = u_l(R_3) = \dot{R}_3. \quad (1.7)$$

Here, σ is the surface tension coefficient, τ denote the normal stresses, and J is the intensity of phase transitions. For vapor, liquid droplet, viscoelastic shell, and carrying liquid the normal stresses can be specified in the form [18, 19], respectively:

$$\begin{aligned} \tau_v &= 2\mu_v \frac{\partial u_v}{\partial r}, \quad \tau_e = 2\mu_e \frac{\partial u_e}{\partial r}, \\ \tau_s &= -\frac{4R_2^2}{r^3} \left(\frac{G_s R_2}{3} \left(1 - \frac{R_{20}^3}{R_2^3} \right) + \mu_s u_s \right), \quad \tau_l = -\frac{4R_3^2}{r^3} \left(\frac{G_l R_3}{3} \left(1 - \frac{R_{30}^3}{R_3^3} \right) + \mu_l u_l \right), \end{aligned} \quad (1.8)$$

where μ is the dynamic viscosity, G is the shear modulus, and the subscript 0 denotes the initial radius in the state of rest. It should be noted that viscoelasticity of the shell and carrying liquid is taken into account in accordance with the rheological Kelvin–Voigt model. Since the vapor viscosity is lower than the liquid viscosity, therefore $\tau_v \ll \tau_e$ and, consequently, in what follows the quantity τ_v can be neglected. The momentum equation (1.3) is integrated from R_1 to infinity. In this case, the interval from R_1 to R_2 will correspond to the liquid droplet parameters, the interval from R_2 to R_3 to the parameters of the viscoelastic shell of droplet, and the interval from R_3 to infinity to the carrying liquid parameters. Then, taking into account (1.4)–(1.8), we can obtain the following equation of radial oscillations of the inclusion considered:

$$\begin{aligned} \rho_d R_1 \dot{U}_e - 0.5 \rho_{d2} U_e^2 + 2 \rho_d \dot{R}_1 U_e &= p_v - p_\infty - \frac{2\sigma_1}{R_1} - \frac{2\sigma_2}{R_2} - \frac{2\sigma_3}{R_3} + \left(\frac{1}{\rho_v} - \frac{1}{\rho_e} \right) J^2 \\ &- \frac{4}{R_1} \mu_d U_e - \frac{4G_s}{3} \left(1 - \frac{R_2^3}{R_3^3} \right) \left(1 - \frac{R_{20}^3}{R_2^3} \right) - \frac{4G_l}{3} \left(1 - \frac{R_{30}^3}{R_3^3} \right), \quad U_e = \dot{R}_1 - \frac{J}{\rho_e}, \\ \rho_d &= \rho_e + (\rho_s - \rho_e) \frac{R_1}{R_2} + (\rho_l - \rho_s) \frac{R_1}{R_3}, \quad \rho_{d2} = \rho_e + (\rho_s - \rho_e) \frac{R_1^4}{R_2^4} + (\rho_l - \rho_s) \frac{R_1^4}{R_3^4}, \\ \mu_d &= \mu_e \left(1 - \frac{R_1^3}{R_2^3} \right) + \mu_s \left(\frac{R_1^3}{R_2^3} - \frac{R_1^3}{R_3^3} \right) + \mu_l \frac{R_1^3}{R_3^3}. \end{aligned} \quad (1.9)$$

We will linearize Eq. (1.9); substituting $\psi = \psi_0 + \psi'$, $\psi' \ll \psi_0$, where $\psi = R_1, R_2, R_3, p$, and J ; we obtain

$$\begin{aligned} \rho_{d0} R_{10} \ddot{R}_1' - \frac{\rho_{d0} R_{10}}{\rho_e} j' + \frac{4\mu_{d0}}{R_{10}} \dot{R}_1' - \frac{4\mu_{d0}}{\rho_e R_{10}} J' &= p_v' - p_\infty' + \frac{2\sigma_1}{R_{10}^2} R_1' + \frac{2\sigma_2}{R_{20}^2} R_2' + \frac{2\sigma_3}{R_{30}^2} R_3' \\ &- \frac{4G_s}{R_{20}} \left(1 - \frac{R_{20}^3}{R_{30}^3} \right) R_2' - \frac{4G_l}{R_{30}} R_3', \quad R_{30}^2 \dot{R}_3' = R_{20}^2 \dot{R}_2' = R_{10}^2 \left(\dot{R}_1' - \frac{J'}{\rho_e} \right). \end{aligned} \quad (1.10)$$

Here, the subscript “0” relates to the parameters of the equilibrium state. When there are no phase transitions $J = J' = 0$, equations (1.9) and (1.10) take the form given in [20].

2. INTERFACIAL MASS TRANSFER

In order to take into account interfacial heat and mass transfer, we will use the heat conduction equations for vapor, liquid phase, and shell [18, 19]

$$\rho_v c_{pv} \frac{\partial T_v}{\partial t} = \frac{1}{r^2} \frac{\partial}{\partial r} \left(\lambda_v r^2 \frac{\partial T_v}{\partial r} \right) + \frac{\partial p_v}{\partial t}, \quad 0 < r < R_1, \quad (2.1)$$

$$\rho_e c_{pe} \frac{\partial T_e}{\partial t} = \lambda_e \frac{\partial^2 T_e}{\partial r^2}, \quad R_1 < r < R_2, \quad (2.2)$$

$$\rho_s c_{ps} \frac{\partial T_s}{\partial t} = \lambda_s \frac{\partial^2 T_s}{\partial r^2}, \quad R_2 < r < R_3, \quad (2.3)$$

where T is the temperature, c_p is the specific heat, and λ is the thermal conductivity coefficient. Here and in what follow, we will assume that the thickness of the internal layer of liquid phase and shell of a droplet is much less than the dimension of a vapor bubble; therefore, the heat conduction equations (2.2) and (2.3) are written in the plane-one-dimensional approximation. In the absence of the outer heat supply, the mass-average temperature of carrying liquid will be assumed to be constant $T_l = T_0 = \text{const}$. In Eq. (2.1) the pressure p_v is considered to be a function of time only: $p_v(t)$; thereby, the homobaricity condition [18] is fulfilled. This assumption makes it possible to avoid solution of the equation of motion of vapor inside the bubble and thereby to simplify the system of equations. In formulation of the problem it is necessary to specify the boundary conditions; for temperatures they take the form:

$$r = 0: \frac{\partial T_v}{\partial r} = 0, \quad r = R_1: T_v = T_e = T_a, \quad J = \frac{1}{L} \left(\lambda_e \frac{\partial T_e}{\partial r} - \lambda_v \frac{\partial T_v}{\partial r} \right), \quad (2.4)$$

$$r = R_2: T_e = T_s, \quad \lambda_e \frac{\partial T_e}{\partial r} = \lambda_s \frac{\partial T_s}{\partial r}, \quad r = R_3: T_s = T_l = T_0 = \text{const}. \quad (2.5)$$

It should be noted that we have adopted here the quasi-equilibrium phase transition scheme; this fact is testified by the condition of equality of the temperature on the surface of a vapor bubble to the saturation temperature T_a . Thus, the phase-transition intensity J can be determined in accordance with the last of relations (2.4) from solution of the heat conduction equations (2.2) and (2.1) for liquid and vapor, respectively. The saturation temperature T_a can be determined from the Clausius–Clapeyron equation

$$T_a = \left(\frac{1}{T_{a0}} - \frac{B_v}{L} \ln \left(\frac{p_v}{p_\infty} \right) \right)^{-1}. \quad (2.6)$$

Here, T_{a0} is the temperature on saturation line that depends on the kind of liquid, B_v is the reduced vapor constant, and L is the specific vaporization heat of liquid. In the case of small oscillations of inclusion with a small amplitude the solution of the above problem (2.1)–(2.6) can be determined in form of the real parts of the following complex-valued functions [21]

$$T_j = T_0 + T'_j \exp(i\omega t), \quad j = v, e, s, \quad p_v = p_\infty + p'_v \exp(i\omega t), \quad (2.7)$$

where ω is the perturbation frequency. Taking into account the solution of form (2.7), we can write Eq. (2.1) as follows:

$$\frac{\partial^2 T'_v}{\partial r^2} + \frac{2}{r} \frac{\partial T'_v}{\partial r} - \frac{i\omega}{\kappa_v} T'_v = -\frac{i\omega}{\lambda_v} p'_v, \quad \kappa_v = \frac{\lambda_v}{\rho_v c_{pv}}. \quad (2.8)$$

The general solution of the second-order differential equation (2.8) can be found in the form of the particular solution $T'_v = \kappa_v p'_v / \lambda_v$ and the linear combination of two independent solutions of the homogeneous equation

$$T'_v = \frac{\kappa_v}{\lambda_v} \left(1 + V_1 \frac{\sinh(rY_v)}{r} + V_2 \frac{\cosh(rY_v)}{r} \right), \quad Y_v = \sqrt{\frac{i\omega}{\kappa_v}},$$

where V_1 and V_2 are the unknown variables. Similarly, the solutions of Eqs. (2.2) and (2.3) can be written as follows:

$$T'_e = E_1 \sinh[(r - R_{10})Y_e] + E_2 \cosh[(r - R_{10})Y_e], \quad Y_e = \sqrt{i\omega/\kappa_e},$$

$$T'_s = S_1 \sinh[(r - R_{20})Y_s] + S_2 \cosh[(r - R_{20})Y_s], \quad Y_s = \sqrt{i\omega/\kappa_s}.$$

The unknown variables V_1 , V_2 , E_1 , E_2 , S_1 , and S_2 can be determined from the boundary conditions (2.4)–(2.6). Thus, we can determine the analytical expressions for the temperature distributions of vapor, liquid droplet, and shell T_v , T_e , and T_s , respectively, and the intensity of phase transition J . In the next stage, the energy equation for a vapor bubble is derived. As the equation of state of vapor, we use the Mendelev–Clapeyron equation

$$p_v = \rho_v B_v T_v = \frac{\gamma - 1}{\gamma} c_{pv} \rho_v T_v, \quad (2.9)$$

where γ is the specific heat ratio. Equation (2.9) can be differentiated with respect to time t and, with regard to (1.1), it takes the form:

$$\frac{dp_v}{dt} = \frac{\gamma-1}{\gamma} \rho_v c_{pv} \frac{dT_v}{dt} - \frac{2}{r} p_v u_v. \quad (2.10)$$

The heat influx equation for the vapor phase (2.1) can be integrated with respect to the radial coordinate from zero to R_{10} , then linearized, and, with regard to (2.10), written as follows:

$$\frac{dp'_v}{dt} = -\frac{3\gamma p_{v0}}{R_{10}} U'_v + 3(\gamma-1) \frac{\lambda_v}{R_{10}} \left(\frac{\partial T'_v}{\partial r} \right)_{r=R_{10}}, \quad U'_v = \dot{R}'_1 - J' / \rho_v.$$

Finally, with regard to the obtained expressions for T'_v and J' , this equation takes the form:

$$\begin{aligned} \frac{dp'_v}{dt} &= -\frac{3\gamma p_{v0}}{R_{10}} \frac{dR'_1}{dt} - 3i\omega \Psi p'_v, \\ \Psi &= 3(\gamma-1) \frac{y_v \coth y_v - 1}{y_v^2} \left(1 - \frac{\lambda_v \tau_v}{\kappa_v} \right) \left(1 - \frac{\gamma}{\gamma-1} \frac{p_{v0}}{\rho_v L} \right) - 3i \frac{\gamma p_{v0}}{\rho_v R_{10}} \frac{\lambda_e y_e}{L r_e} \frac{W \tau_v}{\omega}, \\ \tau_v &= \frac{B_v}{p_{v0} L} \left(\frac{1}{T_{a0}} - \frac{B_v}{L} \ln \left(\frac{p_{v0}}{p_\infty} \right) \right)^{-2}, \quad W = \frac{1 + \chi \coth y_s \coth y_e}{\coth y_e + \chi \coth y_s}, \quad \chi = \frac{\lambda_s y_s r_e}{\lambda_e y_e r_s}, \\ y_v &= \sqrt{\frac{i\omega R_{10}^2}{\kappa_v}}, \quad y_e = \sqrt{\frac{i\omega r_e^2}{\kappa_e}}, \quad y_s = \sqrt{\frac{i\omega r_s^2}{\kappa_s}}, \quad r_e = R_{20} - R_{10}, \quad r_s = R_{30} - R_{20}. \end{aligned} \quad (2.11)$$

Here, r_e is the thickness of the internal layer of liquid phase and r_s is the shell thickness.

3. DISPERSION RELATION

We will represent the solutions of Eqs. (1.10) and (2.11) in the form of traveling waves $R'_j = \Delta R_j \exp(i\omega t)$, $j = 1, 2, 3$, $p'_v = \Delta p_v \exp(i\omega t)$, and $J' = \Delta J \exp(i\omega t)$, where ΔR_j , ΔJ , and Δp_v are the wave amplitudes. We can determine ΔR_1 from the system of equations obtained:

$$\begin{aligned} \Delta R_1 &= -\frac{\Delta p_\infty}{\rho_{d0} R_{10} (\omega_0^2 + i\omega\delta - \omega^2)}, \quad (3.1) \\ \omega_0^2 &= X + \frac{p_{v0}}{\rho_{d0} R_{10}^2} \operatorname{Re} \Phi + \frac{\omega \operatorname{Im} J_0}{\rho_e} - \frac{4\mu_{d0}}{\rho_{d0} \rho_e} \frac{\operatorname{Re} J_0}{R_{10}^2} - \frac{\operatorname{Im} J_0}{\omega \rho_e} \left(X + \frac{2\sigma_1}{\rho_{d0} R_{10}^3} \right), \\ \delta &= \frac{4\mu_{d0}}{\rho_{d0} R_{10}^2} + \frac{p_{v0}}{\rho_{d0} R_{10}^2} \frac{\operatorname{Im} \Phi}{\omega} - \frac{\operatorname{Re} J_0}{\rho_e} - \frac{4\mu_{d0}}{\rho_{d0} \rho_e} \frac{\operatorname{Im} J_0}{\omega R_{10}^2} + \frac{\operatorname{Re} J_0}{\rho_e \omega^2} \left(X + \frac{2\sigma_1}{\rho_{d0} R_{10}^3} \right), \\ J_0 &= \frac{p_{v0} \Phi}{L R_{10}} \left(\frac{\lambda_e y_e}{r_e} W \tau_v - i\omega R_{10} \frac{y_v \coth y_v - 1}{y_v^2} \left(1 - \frac{\lambda_v \tau_v}{\kappa_v} \right) \right), \quad \Phi = \frac{3\gamma}{1 + \Psi}, \\ X &= \frac{1}{\rho_{d0} R_{10}^2} \left(\frac{4G_s R_{10}^3}{R_{20}^3} \left(1 - \frac{R_{20}^3}{R_{30}^3} \right) + \frac{4G_l R_{10}^3}{R_{30}^3} - \frac{2\sigma_1}{R_{10}} - \frac{2\sigma_2 R_{10}^3}{R_{20}^4} - \frac{2\sigma_3 R_{10}^3}{R_{30}^4} \right), \\ p_{v0} &= p_\infty + \frac{2\sigma_1}{R_{10}} + \frac{2\sigma_2}{R_{20}} + \frac{2\sigma_3}{R_{30}}. \end{aligned} \quad (3.2)$$

It should be noted that if the phase transitions are neglected $J_0 = 0$, then the form of the expressions for damping of radial oscillations δ and the resonance frequency of inclusion ω_0 coincides with the corresponding expressions obtained in [20, 22]. When there is only a vapor bubble in liquid, the expressions for δ and ω_0 take the form [23]. We will consider propagation of the acoustic wave through a viscoelastic medium in which there is a variety of liquid droplets covered by shells and located randomly over the wavelength so that inside the droplet there are vapor bubbles. The distances between the inclusions are

Table 1. Thermophysical parameters

	c_{pv} , J/kg K	$\mu_v \times 10^{-6}$, Pa s	B_v	γ	$\lambda_v \times 10^{-2}$, kg m/s ³ K	$L \times 10^5$, m ² /s ²	T_{a0} , K	ρ , kg/m ³	$\mu_e \times 10^{-4}$, Pa s	c_{pe} , J/kg K	$\lambda_e \times 10^{-2}$, kg m/s ³ K
DDFP	842	2.37	28.8	1.05	2	1.16	302	1590	6.52	1090	5.6
OFP	1110	2.3	44.2	1.04	1.38	2.43	236.4	1600	1.08	1270	3.85

assumed to be much greater than their dimensions. As noted above, the dimensions of the internal liquid layer and the droplet shell must satisfy the conditions $r_e \ll R_{l0}$ and $r_s \ll R_{l0}$. In this case, taking the multiple wave scattering into account can be carried out in the way usual for the bubbly medium based on the well-known wave equation of Foldy type in the longwave approximation [24]:

$$\frac{1}{C_l^2} \frac{\partial^2 p_\infty}{\partial t^2} - \frac{\partial^2 p_\infty}{\partial x^2} = 4\pi\rho_l \int_0^\infty R_{l0}^2 \ddot{R}_l f dR_{l0}. \quad (3.3)$$

Here, f is the inclusion-size distribution function and C_l is the speed of sound in carrying liquid. Taking into account the fact that $\ddot{R}_l = -\omega^2 \Delta R_l$ and $\partial^2 p_\infty / \partial t^2 = -\omega^2 \Delta p_\infty$ and with regard to expression (3.1), the wave equation (3.3) can be written in the form:

$$\frac{\partial^2 \Delta p_\infty}{\partial x^2} + K_*^2 \Delta p_\infty = 0, \quad \frac{K_*^2}{\omega^2} = \frac{1}{C_l^2} + 4\pi\rho_l \int_0^\infty \frac{R_{l0} f}{\rho_{d0}(\omega_0^2 + i\omega\delta - \omega^2)} dR_{l0}, \quad (3.4)$$

where K_* is the complex wavenumber and the second of Eqs. (3.4) is the dispersion relation. If we take $f = n\delta(R - R_{l0})$, where δ is the Dirac function and $n = 3\alpha/4\pi R_{l0}^3$ is the number of inclusions per unit liquid volume, then integral on (3.4) can be omitted and the inclusions are considered to be monodisperse.

4. CALCULATION RESULTS

In Fig. 2 we have reproduced the results of comparison of the dependences of the phase velocity $C_p = \omega / \text{Re } K_*$ (Fig. 2a) and the attenuation coefficient $K_{**} = \text{Im } K_*$ (Fig. 2b) for a mixture of water with dodecafluoropentane C_5F_{12} (DDFP) and vapor bubbles at the center (curve 1) and octafluoropropane C_3F_8 (OFP) droplets and vapor bubbles at the center (curve 2). It should be noted that the dispersion curves are almost the same for two different matters DDFP and OFP without regard for phase transitions; therefore, in Fig. 2 we have plotted a single curve 3. The mixture temperature $T_0 = 310$ K, $p_\infty = 10^5$ Pa, the volumetric inclusion contain $\alpha = 0.001$, the vapor bubble radius $R_{l0} = 7 \times 10^{-6}$ m, and the liquid droplet thickness $r_e = 10^{-6}$ m. In this stage the elastic shell of droplets is neglected; therefore, $r_s = 0$. The calculations were carried out using the dispersion relation (3.4). The thermophysical parameters of the matters DDFP and OFP are borrowed from [13, 25] and given in Table 1. As can be seen from Fig. 2, the presence of phase transitions leads to a significant decrease in the phase velocity (Fig. 2a) and increase in the attenuation coefficient (Fig. 2b) at the frequencies lower than the resonance inclusion frequency. In this case the resonance inclusion frequency itself also depends on the intensity of phase transitions. This also follows from formula (3.2). For the DDFP droplets the intensity of mass exchange is higher than that for the OFP droplets. The fact is testified by the lower speed of sound and the higher attenuation coefficient. This result is in agreement with conclusions [5–7]: to evaporate a DDFP droplet it is necessary to expend more acoustic energy as compared with an OFP droplet. Note that in a particular case a similar pattern is observed for the mixture of water with steam bubbles [23, 26]. Taking the mass exchange processes into account leads to decrease in the bubble resonance frequency and the phase velocity, as well as to increase in the attenuation coefficient at the subresonance frequencies.

In Fig. 3 we have reproduced the plots of the phase velocity (Figs. 3a and 3c) and the attenuation coefficient (Figs. 3b and 3d) for a mixture of water with DDFP droplets covered by polymeric shell and vapor bubbles at the center for various shell dimensions r_s and shear moduli G_s . The shell parameters are as follows; $\rho_s = 1475$ kg/m³ and $\mu_s = 0.99$ Pa s. The remaining parameters are the same as those in Fig. 2. When there is no elastic droplet shell $r_s = 0$ and the surface tensions and the elasticity of carrying

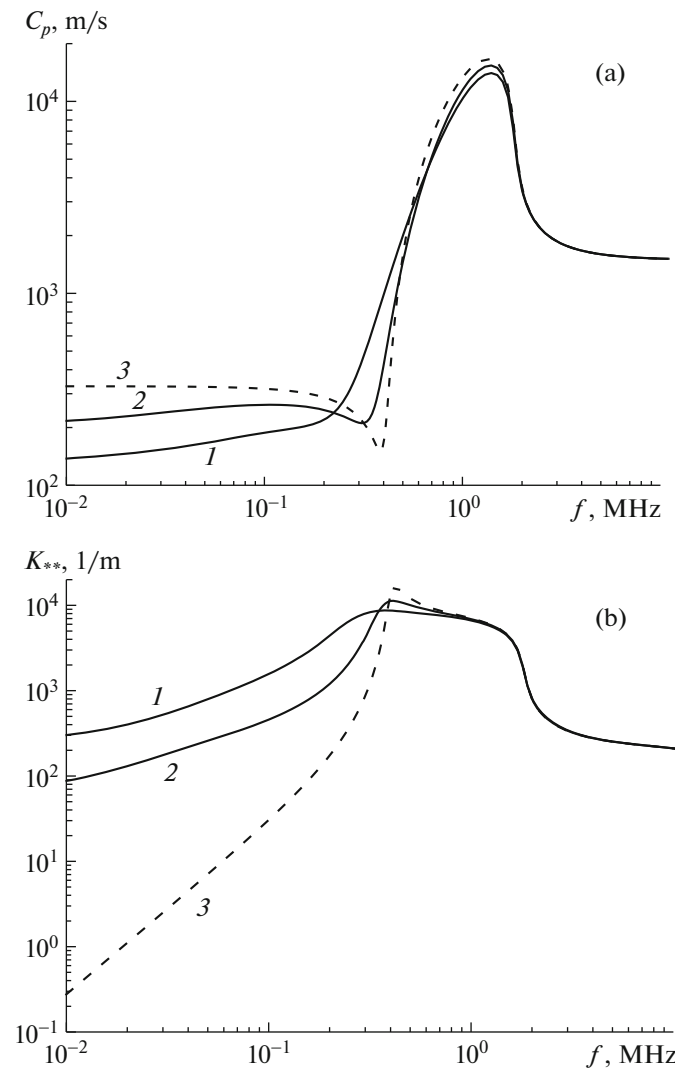


Fig. 2. Phase velocity (a) and attenuation coefficient (b) as functions of the perturbation frequency for the mixture of water with liquid droplets and vapor bubbles at the center: curves 1 corresponds to DDFP, curves 2 to OFP, and curves 3 are constructed without regard for mass exchange.

phase $G_l = 0$ are neglected, then from formula (3.2) we can obtain for the resonance inclusion frequency the following expression

$$f_r = \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \left(\frac{p_\infty}{\rho_{d0} R_{l0}^2} \operatorname{Re} \Phi + \frac{\omega \operatorname{Im} J_0}{\rho_e} - \frac{4\mu_{d0}}{\rho_{d0} \rho_e} \frac{\operatorname{Re} J_0}{R_{l0}^2} \right)^{\frac{1}{2}}.$$

Thus, from Fig. 3 it follows that increase in the thickness and shear modulus of the elastic shell of droplet leads to decrease in the attenuation coefficient of acoustic wave and increase in the phase velocity at frequencies $f < f_r$. In this case the inclusion resonance frequency itself increases and the range of variation in the phase velocity decreases. Calculations also showed that with increase in the shear modulus of carrying liquid G_l the dispersion curves obtained coincide with curves in Figs. 3c and 3d; for this reason, they are omitted here. The above statements are in good agreement with experiments [27, 28] for an elastic medium with gas bubbles and for water with gas bubbles covered by rubber shell. The theory was compared with the experimental data in authors' studies [29, 30]. Note that in these experiments the bubble dimensions were of the order of 1 mm and 1 cm. In [31, 32] the measurements of the attenuation coefficient in a mixture of liquid with encapsulated bubbles of the Sonazoid type which are closer to medical applications were given. The bubbles were covered by a polymeric material and filled with a fluorocarbon gas.

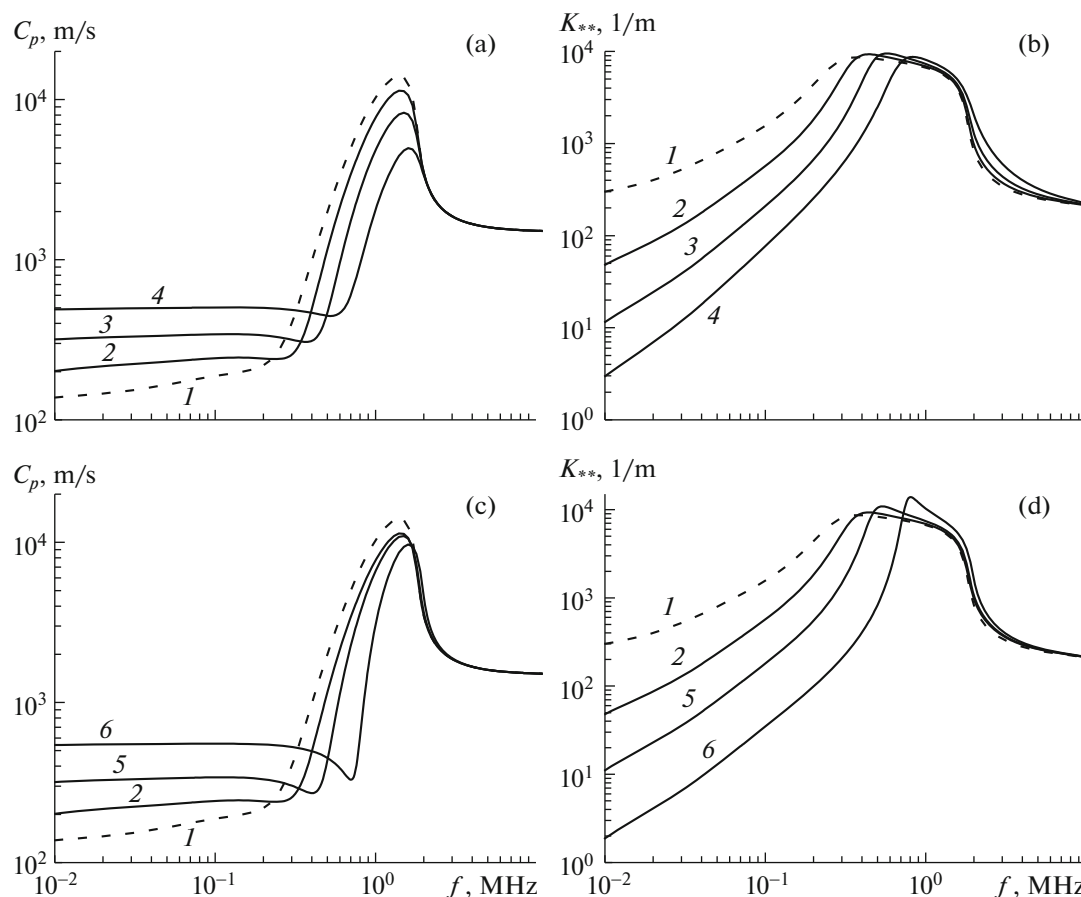


Fig. 3. Phase velocity (a and c) and attenuation coefficient (b and d) for various shell thicknesses r_s (a and b) and shear moduli G_s (c and d): curves 1–4 correspond to $r_s = 0, 10^{-8}, 3 \times 10^{-8}$, and 8×10^{-8} at $G_s = 10^7$ Pa; curves 5–6 correspond to $G_s = 3 \times 10^7$ and 10^8 Pa at $r_s = 10^{-8}$ m.

The mean bubble dimension was equal to $R_{10} = 2.6 \times 10^{-6}$ m, the number of bubbles per unit liquid volume $n = 2.5 \times 10^{10}$, the shell thickness $r_s = 4 \times 10^{-9}$ m, the viscosity $\mu_s = 0.9$ Pa s, and the shear modulus $G_s = 50 \times 10^6$ Pa. In Fig. 4 we have compared the theory with the data of measurement results. The calculation was carried out from formula (3.4) under the assumption of absence of phase transitions $J_0 = 0$ and liquid interlayer $r_e = 0$. As shown in Fig. 4, the presence of the fairly thin elastic shell of bubbles (the shell thickness is less than the gas bubble radius by 650 times) leads to both the higher resonance frequency and an appreciable decrease in the peak attenuation coefficient. This is confirmed by the experimental data. Physically, this is attributable to follows. The bubble shell or the elastic carrying phase slows down the radial bubble oscillations. The greater the shell thickness or the shear modulus, the lesser the intensity of radial oscillations and, as a result, the acoustic wave is scattered to a lesser extent as compared with the usual bubbly media.

SUMMARY

Dynamics of acoustic waves in the viscoelastic medium with liquid droplets covered by an elastic shell inside which vapor bubbles are located are first considered.

It is found that in propagation of acoustic wave in a mixture of liquid with perfluorocarbon droplets and vapor bubbles at the center phase transitions lead to significant decrease in the phase velocity and increase in the attenuation coefficient at the frequencies lower than the inclusion resonance frequency which also depends on the intensity of phase transitions. Since in considering the dodecafluoropentane

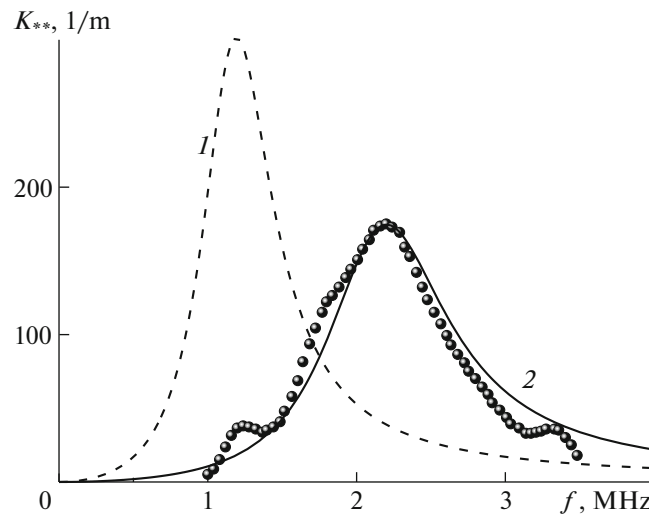


Fig. 4. Comparison of the attenuation coefficient as a function of the perturbation frequency with the experimental data [31]; curves 1–2 correspond to $r_s = 0$ and 4×10^{-9} m.

C_5F_{12} droplets the phase velocity is lower and the wave attenuation coefficient is greater than those for the octafluoropropane C_3F_8 droplets, for this reason the intensity of phase transitions for C_5F_{12} is higher than for C_3F_8 .

It is obtained that increase in the thickness and shear modulus of the elastic shell of droplet and increase in the shear modulus of the elastic carrying medium lead to displacement of the dispersion curves towards increase in the perturbation frequency. In a particular case, for the mixture of liquid with encapsulated gas bubbles it was found that there is a good agreement between the theoretical dependence of the wave attenuation coefficient and the known experimental data. It is shown that the presence of fairly thin elastic shell of droplets leads to an appreciate decrease in the peak attenuation coefficient and the higher bubble resonance frequency.

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